

The Patriot system has four major operational functions: communication, command and control, radar surveillance and missile guidance. These functions combine to provide a coordinated, secure, integrated, mobile air defence system.

Patriot missile is equipped with a TVM guidance system. Target is acquired in the terminal phase of the flight by the target acquisition system in the missile. It transmits data using the TVM downlink via the ground radar to the engagement control station for computation of the final course correction. Course correction commands are transmitted to the missile via the missile track command uplink.

The missile has a range of 70km and maximum altitude greater than 24km. Minimum flight time, which is the time to arm the missile, is less than nine seconds, and the maximum flight time is less than 3½ minutes. It is a supersonic missile with maximum speed of Mach 5.

Different variants of Patriot missile system include MIM-104A, MIM-104B (PAC-1), MIM-104C (PAC-2), MIM-104D (PAC-2/GEM), MIM-104F (PAC-3) and Patriot Advanced Affordable Capability-4 (PAAC-4).

The missile system was inducted into service in 1981 and is in service till date. Other than the USA, a large number of other allied nations, including Egypt, Germany, Greece, Israel, Japan, Kuwait, the Netherlands, Saudi Arabia and Taiwan, have this system in their arsenal.

Patriot missile systems were first used during the Gulf War of 1991. There have been controversies about the success rate of the missile system in engaging ballistic missiles during the Gulf War. The system was again deployed in Iraq in 2003 during Operation Iraqi Freedom. The systems were stationed in Kuwait and were reportedly used with success against hostile ballistic missiles.

Recently, during Israel-Gaza conflict of 2014 called Operation Protective Edge, Patriot missile system was



Fig. 6: MIM-104 Patriot immediately after launch



Fig. 7: RIM-174 Standard ERAM (SM-6) in flight

successfully used to bring down two unmanned aerial vehicle drones of Hamas. Fig. 6 shows Patriot missile launch.

AIM-54 Phoenix, manufactured by Hughes Aircraft Co. and Raytheon Corp. and capable of attacking more than one aircraft with multiple launches, is a supersonic, radar-guided, long-range air-to-air missile. The missile is carried in clusters of up to six missiles, and while in service during 1974 to 2004, it was used by the US Navy and Air Force of Islamic Republic of Iran.

F-14 Tomcat was the only launch

platform capable of carrying AIM-54 Phoenix. The missile system retired from service in 2004 in favour of AIM-120 AMRAAM. This Mach 5 missile has a maximum operational range of 190km and employs a combination of semi-active (mid-course correction) and active radar homing guidance (terminal guidance).

AIM-120 AMRAAM (advanced medium-range air-to-air missile) is a high-supersonic, day/night/all-weather beyond-visual-range (BVR), fire-and-forget air-to-air missile. Manufactured by Hughes from 1991 to 1997 and Raytheon from 1997 till date, it has different variants, which include AIM-120A, AIM-120B, AIM-120C and AIM-120D, with AIM-120D having an operational range in excess of 180km. Operational ranges for other variants are 55km to 75km (AIM-120 A/B) and greater than 105km (AIM-120C).

It employs inertial navigation system (INS) for mid-course guidance and active radar homing for guidance in terminal phase. Once the missile closes to self-homing distance, the active radar guides it towards the target. This feature provides fire-and-forget capability to the missile and allows the pilot to fire a number of

missiles simultaneously at multiple targets. Its capabilities include look-down, shoot-down, multiple launches against multiple targets and intercepts at very short range in dog-fight situations. It was inducted into service in 1991 and is in service till date with the US Air Force and Navy and more than 25 allies of the USA.

RIM-174 Standard extended-range active-missile (ERAM), also called Standard Missile 6 (SM-6), is a long-range anti-air-warfare missile designed for use by the navy against a range of aerial targets including fixed and rotary wing aircraft, un-